Flatness-Based and Passivity-Based Control with Adaptive RBF Neural Compensation of a 2 MW Direct-Drive PMSG Wind Energy Conversion System

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***Abstract*—This paper presents the modeling and control of a 2 MW direct-drive wind energy conversion system based on a permanent magnet synchronous generator (PMSG) connected to the grid through a back-to-back converter and an L filter. The generator is modeled in the receptor convention, and the signs of the complete model are verified by an exact power balance. The rotor speed and the energy stored in the DC link and in the filter are used as flat outputs of the two outer loops, and the currents of both converters are controlled by passivity-based control (PBC). A radial basis function (RBF) neural network, adapted online by a Lyapunov law with σ-modification, compensates the uncertainty of the aerodynamic-torque model and of the generator parameters. Simulations with step and turbulent wind, parameter mismatch and a 50 % grid voltage dip show that all the controlled variables track their references, that the DC-link voltage remains within 1.7 V of 1200 V outside the dip, and that the neural compensation reduces the integral absolute speed error by 27 to 49 %.**

***Index Terms*—PMSG, wind energy conversion, flatness, passivity-based control, RBF neural network, back-to-back converter.**

I. Introduction

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1], [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]. The machine-side converter (MSC) controls the generator torque to track the maximum power point, and the grid-side converter (GSC) regulates the DC-link voltage and injects the power into the grid.

These tasks are usually performed by cascaded PI loops in rotating frames [5], [6]. Differential flatness [7] computes the control directly from the desired trajectory of a flat output and has been applied to the PMSG wind turbine and its DC link [8]; with the stored energy as flat output, the DC-link dynamics become a linear power balance [9]. Passivity-based control [10] shapes the energy of the closed loop and keeps the workless coupling terms of electrical machines. Both methods use a model; their weak point is the aerodynamic torque, which is not measured, and the parameter uncertainty, which neural networks can learn online [11].

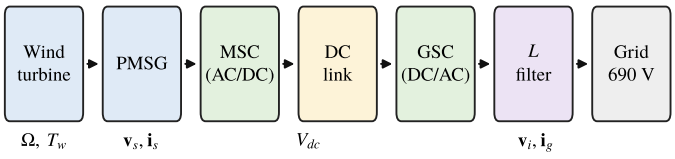


Fig. 1. Direct-drive PMSG wind energy conversion system with a back-to-back converter.

This paper presents:

- a complete model of a 2 MW direct-drive PMSG wind energy conversion system in the receptor convention, with an exact power balance that verifies its signs;
- flat outer loops for the rotor speed and the DC-link energy, and PBC current loops for both converters;
- an RBF network that compensates the aerodynamic-torque and parameter uncertainty in the flat speed law, with a Lyapunov stability proof;
- simulation results without and with the neural compensation under step and turbulent wind, parameter mismatch and a grid voltage dip.

II. System Modeling

The studied system is shown in Fig. 1. The turbine drives the PMSG directly; the MSC rectifies the generator voltages, the DC link decouples the two converters, and the GSC injects the power into the 690 V, 50 Hz grid through an L filter.

A. Wind Speed

Two wind profiles are used. The step profile (9 m/s, 11 m/s at 10 s and 8 m/s at 25 s), similar to the one of [8], assesses the transient response. The turbulent profile is

$$v(t) = V_m + \sum_{k=1}^{3} A_k \sin(2\pi f_k t) + v_t(t) \qquad (1)$$

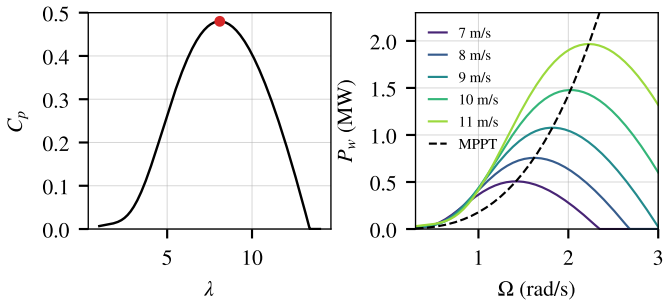


Fig. 2. Power coefficient for $\beta = 0$ (left) and captured power versus rotor speed with the maximum power curve (right).

with $V_m = 8.5$ m/s, $A = [1.5, 0.2, 0.3]$ m/s and $f = [0.03, 0.16, 0.17]$ Hz [15]; the turbulence v_t is a first-order coloured noise with a standard deviation of 0.5 m/s and a correlation time of 1.5 s. Over 40 s, the wind remains between 6.0 and 10.9 m/s, below the rated wind speed, so the pitch angle is zero. The controller uses the anemometer measurement v_m, modeled by a first-order lag of 1 s.

B. Wind Turbine

The power captured by the rotor and the tip-speed ratio are [8], [12]

$$P_w = \tfrac{1}{2}\rho\pi R^2 C_p(\lambda,\beta)\, v^3, \qquad \lambda = \frac{\Omega R}{v} \tag{2}$$

where ρ is the air density, R the rotor radius, Ω the rotor speed and β the pitch angle. The power coefficient is

$$C_p = 0.5176\Big(\frac{116}{\lambda_i} - 0.4\beta - 5\Big)e^{-21/\lambda_i} + 0.0068\lambda, \tag{3}$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$. For $\beta = 0$, $C_{p,\max} = 0.48$ at $\lambda_{opt} = 8.1$ (Fig. 2). The aerodynamic torque is $T_w = P_w/\Omega$. For $P_n = 2$ MW and $R = 40$ m, close to the 2 MW turbine of [14], the rated wind speed given by $P_n = \frac{1}{2}\rho\pi R^2 C_{p,\max} v_n^3$ is $v_n = 11.0$ m/s, the rated speed is $\Omega_n = \lambda_{opt} v_n/R = 2.23$ rad/s and the rated torque is $T_n = P_n/\Omega_n = 0.897$ MN·m.

C. Shaft

The direct-drive shaft is described by the one-mass model

$$J\frac{d\Omega}{dt} = T_w + T_e - B\Omega \tag{4}$$

where J is the total inertia, B the friction coefficient and T_e the electromagnetic torque in the receptor convention ($T_e < 0$ in generating mode).

D. PMSG

In the rotor dq frame (amplitude-invariant transformation, d axis on the magnet flux), with the receptor convention and $L_d = L_q = L_s$ [8],

$$v_{sd} = R_s i_{sd} + L_s\frac{di_{sd}}{dt} - \omega_e L_s i_{sq} \tag{5}$$

$$v_{sq} = R_s i_{sq} + L_s\frac{di_{sq}}{dt} + \omega_e L_s i_{sd} + \omega_e\psi_f \tag{6}$$

$$T_e = \tfrac{3}{2} p\,\psi_f\, i_{sq} \tag{7}$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux. With $\mathbf{i}_s = [i_{sd}\ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd}\ v_{sq}]^T$, $\mathbf{e}_q = [0\ 1]^T$ and the skew-symmetric matrix $\boldsymbol{\mathcal{J}} = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$,

$$L_s\dot{\mathbf{i}}_s = \mathbf{v}_s - R_s\mathbf{i}_s + \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{i}_s - \omega_e\psi_f\mathbf{e}_q. \tag{8}$$

The generator data are those of the non-salient 2 MW PMSG of [13] (Table I): at Ω_n, the back-EMF $p\Omega_n\psi_f = 532$ V is close to the peak phase voltage of the grid (563 V).

E. Back-to-Back Converter and DC Link

The two-level converters are represented by their average model, with voltage vectors limited to $V_{dc}/\sqrt{3}$. The DC link receives the power of the MSC and supplies the GSC [8]:

$$CV_{dc}\frac{dV_{dc}}{dt} = P_{dc} - P_i, \quad P_{dc} = -\tfrac{3}{2}\mathbf{v}_s^T\mathbf{i}_s, \quad P_i = \tfrac{3}{2}\mathbf{v}_i^T\mathbf{i}_g. \tag{9}$$

F. L Filter and Grid

In the grid-voltage-oriented frame ($v_{gq} = 0$, ideal PLL), with the current counted towards the grid [8],

$$L_f\dot{\mathbf{i}}_g = \mathbf{v}_i - R_f\mathbf{i}_g + \omega_g L_f\boldsymbol{\mathcal{J}}\mathbf{i}_g - \mathbf{v}_g, \tag{10}$$

and the grid powers are $P_g = \frac{3}{2}v_{gd}i_{gd}$ and $Q_g = -\frac{3}{2}v_{gd}i_{gq}$.

G. Power Balance

With $E = \frac{1}{2}J\Omega^2 + \frac{3}{4}L_s\|\mathbf{i}_s\|^2 + \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$, and since $\mathbf{x}^T\boldsymbol{\mathcal{J}}\mathbf{x} = 0$ and $T_e\Omega = \frac{3}{2}\omega_e\psi_f i_{sq}$, the model gives, for any state and converter voltages,

$$\dot{E} = P_w - B\Omega^2 - \tfrac{3}{2}R_s\|\mathbf{i}_s\|^2 - \tfrac{3}{2}R_f\|\mathbf{i}_g\|^2 - P_g, \tag{11}$$

which verifies the sign of every term; its numerical residual is below $10^{-15}P_n$.

III. Control

The control structure is shown in Fig. 3. All errors are defined as measured minus reference values. The controller is sampled at $T_s = 1/(2f_{sw}) = 231.5$ μs, and the converter voltages are applied one sample after their computation.

A. MPPT Reference

The optimal speed $\Omega_{opt} = \lambda_{opt} v_m/R$, limited to $[0.5\Omega_n, \Omega_n]$, is filtered by $\ddot{\Omega}^* = \omega_r^2(\Omega_{opt} - \Omega^*) - 2\omega_r\dot{\Omega}^*$ ($\omega_r = 1$ rad/s), which gives a smooth reference Ω^* and its derivative.

TABLE I
PARAMETERS OF THE STUDIED SYSTEM

Parameter	Symbol	Value	Source
Rated power	P_n	2 MW	[13]
Rotor radius	R	40 m	[14]
Air density	ρ	1.225 kg/m^3	[8]
Rated wind / rotor speed	v_n, Ω_n	11.0 m/s, 2.23 rad/s	(2)
Total inertia	J	2.8×10^6 kg·m^2	assumed
Friction coefficient	B	10^3 N·m·s	assumed
Pole pairs	p	26	[13]
Stator resistance	R_s	0.78 mΩ	[13]
Stator inductance	L_s	1.57 mH	[13]
Magnet flux	ψ_f	9.18 Wb	[13]
DC link	C, V_{dc}^*	20 mF, 1200 V	[13]
Filter	L_f, R_f	0.15 mH, 2 mΩ	[13], assumed
Grid		690 V, 50 Hz	[13]
Switching frequency	f_{sw}	2160 Hz	[13]

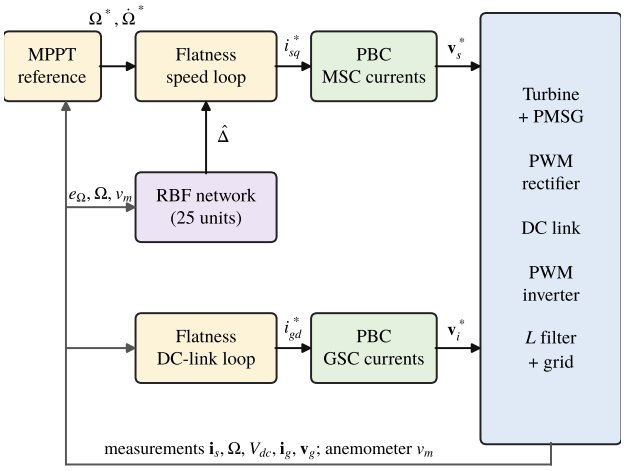


Fig. 3. Control structure: flat outer loops, PBC current loops and RBF compensation.

B. Flatness-Based Speed Loop

From (4), $T_e = J\dot{\Omega} + B\Omega - T_w$: Ω is a flat output of the shaft with input T_e, i.e., i_{sq} [7], [8]. Imposing $\dot{\Omega} = \nu_1$ with $\nu_1 = \dot{\Omega}^* - k_{11}e_\Omega - k_{12}\int e_\Omega dt$, $e_\Omega = \Omega - \Omega^*$, gives

$$T_e^* = J\nu_1 + B\Omega - \hat{T}_w - \hat{\Delta}, \qquad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f}, \quad i_{sd}^* = 0 \tag{12}$$

where $\hat{T}_w$ is computed from (2)–(3) with v_m, and $\hat{\Delta}$ is the neural compensation ($\hat{\Delta} = 0$ without the network). T_e^* is limited to $[-1.2T_n, 0]$, with the integration frozen at the limit. With $i_{sq} = i_{sq}^*$ and $\Delta = T_w - \hat{T}_w$,

$$J\big(\dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega dt\big) = \Delta - \hat{\Delta}. \tag{13}$$

C. Flatness-Based DC-Link Loop

The flat output is the energy of the DC link and of the filter, $y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$. From (9)–(10),

$$\dot{y}_2 = P_{dc} - P_g - \tfrac{3}{2}R_f\|\mathbf{i}_g\|^2, \tag{14}$$

which is linear in the grid power [9]. Imposing $\dot{y}_2 = -k_{21}e_y - k_{22}\int e_y\,dt$, with $e_y = y_2 - y_2^*$ and $y_2^* = \frac{1}{2}CV_{dc}^{*2} + \frac{3}{4}L_f\|\mathbf{i}_g^*\|^2$, gives

$$P_g^* = P_{dc} - \tfrac{3}{2}R_f\|\mathbf{i}_g\|^2 + k_{21}e_y + k_{22}\int e_y dt, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}} \tag{15}$$

and $i_{gq}^* = 0$ (unity power factor). P_{dc} is computed from the applied MSC voltages and the measured currents, so the generator power is passed to the grid before it changes V_{dc}.

D. Passivity-Based Current Loops

With $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$ and $\dot{\mathbf{z}}_s = \mathbf{e}_s$, the MSC voltage reference is

$$\mathbf{v}_s^* = L_s\dot{\mathbf{i}}_s^* + R_s\mathbf{i}_s^* - \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{i}_s^* + \omega_e\psi_f\mathbf{e}_q - R_a\mathbf{e}_s - K_i\mathbf{z}_s \tag{16}$$

and, with $\mathbf{e}_g = \mathbf{i}_g - \mathbf{i}_g^*$ and $\dot{\mathbf{z}}_g = \mathbf{e}_g$, the GSC voltage reference is

$$\mathbf{v}_i^* = L_f\dot{\mathbf{i}}_g^* + R_f\mathbf{i}_g^* - \omega_g L_f\boldsymbol{\mathcal{J}}\mathbf{i}_g^* + \mathbf{v}_g - R_b\mathbf{e}_g - K_{ig}\mathbf{z}_g. \tag{17}$$

The reference derivatives are obtained by first-order filters (2 ms), and the integrals are frozen when the voltage reaches $V_{dc}/\sqrt{3}$.

Proposition 1. *With (16) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2}L_s\|\mathbf{e}_s\|^2 + \frac{1}{2}K_i\|\mathbf{z}_s\|^2$ satisfies $\dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2$ and $\mathbf{e}_s \to 0$. The same holds for (17) applied to (10).*

Proof. Subtracting $L_s\dot{\mathbf{i}}_s^*$ from (8) and inserting (16) gives $L_s\dot{\mathbf{e}}_s = -(R_s + R_a)\mathbf{e}_s + \omega_e L_s\boldsymbol{\mathcal{J}}\mathbf{e}_s - K_i\mathbf{z}_s$. Hence $\dot{H} = \mathbf{e}_s^T L_s\dot{\mathbf{e}}_s + K_i\mathbf{z}_s^T\mathbf{e}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2$, since $\mathbf{e}_s^T\boldsymbol{\mathcal{J}}\mathbf{e}_s = 0$. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [16] gives $\mathbf{e}_s \to 0$. □

E. Adaptive RBF Compensation

The uncertainty $\Delta = T_w - \hat{T}_w$, due to the C_p model, the anemometer and the generator parameters, is approximated on the operating region by $\Delta = \mathbf{W}^{*T}\boldsymbol{\phi}(\mathbf{z}) + \varepsilon$, $|\varepsilon| \le \bar{\varepsilon}$, with $\mathbf{z} = [\Omega/\Omega_n,\ v_m/v_n]^T$ and 25 Gaussian functions $\phi_j = \exp(-\|\mathbf{z} - \mathbf{c}_j\|^2/(2b^2))$ centred on a 5×5 grid of $[0.5, 1.1]^2$, $b = 0.15$. The compensation $\hat{\Delta} = \hat{\mathbf{W}}^T\boldsymbol{\phi}$ replaces the integral of the speed loop ($k_{12} = 0$ when the network is used) and is adapted by

$$\dot{\hat{\mathbf{W}}} = \Gamma\big(\boldsymbol{\phi}\, e_\Omega - \sigma\hat{\mathbf{W}}\big), \tag{18}$$

frozen when T_e^* saturates, with $\hat{\mathbf{W}}(0) = 0$.

Proposition 2. *Consider (13) with $k_{12} = 0$ and the law (18) with $\Gamma, \sigma > 0$, exact J and B, and no saturation. Then e_Ω and $\tilde{\mathbf{W}} = \mathbf{W}^* - \hat{\mathbf{W}}$ are uniformly ultimately bounded.*

Proof. With $V = \frac{1}{2}Je_\Omega^2 + \frac{1}{2\Gamma}\|\tilde{\mathbf{W}}\|^2$ and (13), $\dot{V} = -Jk_{11}e_\Omega^2 + e_\Omega(\tilde{\mathbf{W}}^T\boldsymbol{\phi} + \varepsilon) - \frac{1}{\Gamma}\tilde{\mathbf{W}}^T\dot{\hat{\mathbf{W}}} = -Jk_{11}e_\Omega^2 + e_\Omega\varepsilon + \sigma\tilde{\mathbf{W}}^T\hat{\mathbf{W}}$. Using $\tilde{\mathbf{W}}^T\hat{\mathbf{W}} \le \frac{1}{2}\|\mathbf{W}^*\|^2 - \frac{1}{2}\|\tilde{\mathbf{W}}\|^2$ and $e_\Omega\varepsilon \le \frac{Jk_{11}}{2}e_\Omega^2 + \frac{\bar{\varepsilon}^2}{2Jk_{11}}$ gives $\dot{V} \le -\frac{Jk_{11}}{2}e_\Omega^2 - \frac{\sigma}{2}\|\tilde{\mathbf{W}}\|^2 + \frac{\bar{\varepsilon}^2}{2Jk_{11}} + \frac{\sigma}{2}\|\mathbf{W}^*\|^2$, which is negative outside a compact set. □

TABLE II
CONTROLLER PARAMETERS

Loop	Parameters
Speed (flat)	$\omega_\Omega = 2$ rad/s ($k_{11} = 4$, $k_{12} = 4$), $\omega_r = 1$ rad/s
DC link (flat)	$\omega_v = 60$ rad/s ($k_{21} = 120$, $k_{22} = 3600$)
PBC MSC	$\omega_i = 500$ rad/s: $R_a = 1.57\ \Omega$, $K_i = 393\ \Omega$/s
PBC GSC	$\omega_i = 500$ rad/s: $R_b = 0.148\ \Omega$, $K_{ig} = 37.5\ \Omega$/s
RBF network	25 units, $b = 0.15$, $\Gamma = 10^7$, $\sigma = 0.02\ \mathrm{s}^{-1}$

The gains place double closed-loop poles: $k_{11} = 2\omega_\Omega$, $k_{12} = \omega_\Omega^2$, $k_{21} = 2\omega_v$, $k_{22} = \omega_v^2$, $R_s + R_a = 2\omega_i L_s$, $K_i = \omega_i^2 L_s$, $R_f + R_b = 2\omega_i L_f$, $K_{ig} = \omega_i^2 L_f$ (Table II).

IV. SIMULATION RESULTS

A. Setup

The model is integrated with a fourth-order Runge–Kutta scheme at the step T_s over 40 s, starting from the MPPT equilibrium. Three cases are considered: (i) step wind with nominal parameters; (ii) turbulent wind with nominal parameters; (iii) robustness: turbulent wind, true plant with $C_p \times 1.1$, $J \times 1.2$, $R_s \times 1.5$, $L_s \times 1.2$, $\psi_f \times 0.95$, $R_f \times 1.5$, $L_f \times 1.25$, and a 50 % grid voltage dip between 25 and 25.15 s. The proposed control is evaluated without the network ($\hat{\Delta} = 0$, integral action in the speed loop) and with the network.

B. Step Wind

Fig. 4 shows that all the controlled variables follow their references. The rotor speed follows the MPPT reference from 1.82 to 2.23 and 1.62 rad/s; i_{sq} and i_{gd} follow their references during the transients, i_{sd} and i_{gq} remain at zero, and V_{dc} stays within 1.7 V of 1200 V, since the generator power is passed to the grid through P_{dc} in (15).

C. Turbulent Wind and Robustness

Fig. 5 shows the robustness case. The rotor speed follows its reference, and C_p stays close to its maximum, except during the fast wind decrease around 22 s, where the inertia prevents the rotor from slowing down quickly. The MPPT efficiency, i.e., the ratio between the captured energy and the maximum available energy, is 97.7 %.

D. Effect of the Neural Compensation

Table III and Fig. 6 compare the control without and with the network. The network reduces the IAE of the speed by 49 % for the step wind, 27 % for the turbulent wind and 29 % in the robustness case: after each wind step, the speed error vanishes in 2 to 3 s instead of 5 to 7 s. The network estimate follows the actual uncertainty, about 82 kN·m on average in the robustness case (Fig. 7). The current errors and the DC-link deviation are practically unchanged, and the captured energy is the same; the price is a slightly more active torque (11–14 % larger standard deviation of its rate).

TABLE III
RESULTS WITHOUT AND WITH THE RBF NETWORK (40 S, FROM $t = 1$ S)

Metric	Step w/o	Step with	Turbulent w/o	Turbulent with	Robustness w/o	Robustness with
IAE of Ω (rad)	0.074	**0.037**	0.144	**0.105**	0.177	**0.126**
RMS e_{isq} (A)	0.037	0.036	0.043	0.041	0.047	0.044
Max $\lvert V_{dc} - V_{dc}^* \rvert$ (V)[a]	1.70	1.67	0.16	0.16	0.24	0.25
Energy (kWh)	13.86	13.86	10.05	10.04	10.98	10.97
MPPT efficiency (%)	98.0	98.0	97.7	97.7	97.7	97.7
Torque rate std (MN·m/s)	0.218	0.247	0.187	0.207	0.187	0.212

[a] outside the grid voltage dip.

E. Grid Voltage Dip

During the 50 % dip (Fig. 7), the grid current doubles to keep the injected power, and the DC-link voltage deviates by 14 V (1.2 %) before returning to 1200 V within about 50 ms.

These results are obtained with the average converter model and an ideal PLL; the switching ripple and the PLL dynamics are not represented.

V. CONCLUSION

A 2 MW direct-drive PMSG wind energy conversion system was modeled from the wind to the grid in the receptor convention, with an exact power balance that verifies the signs of the model. Its control combines flatness for the speed and DC-link energy loops, passivity-based control for the currents of both converters, and an RBF network adapted by a Lyapunov law that compensates the aerodynamic-torque and parameter uncertainty. In simulation, all the controlled variables track their references, the DC-link voltage remains within 1.7 V of 1200 V outside the grid dip, and the network reduces the speed tracking error by 27 to 49 % at the price of a slightly more active torque, with the same captured energy.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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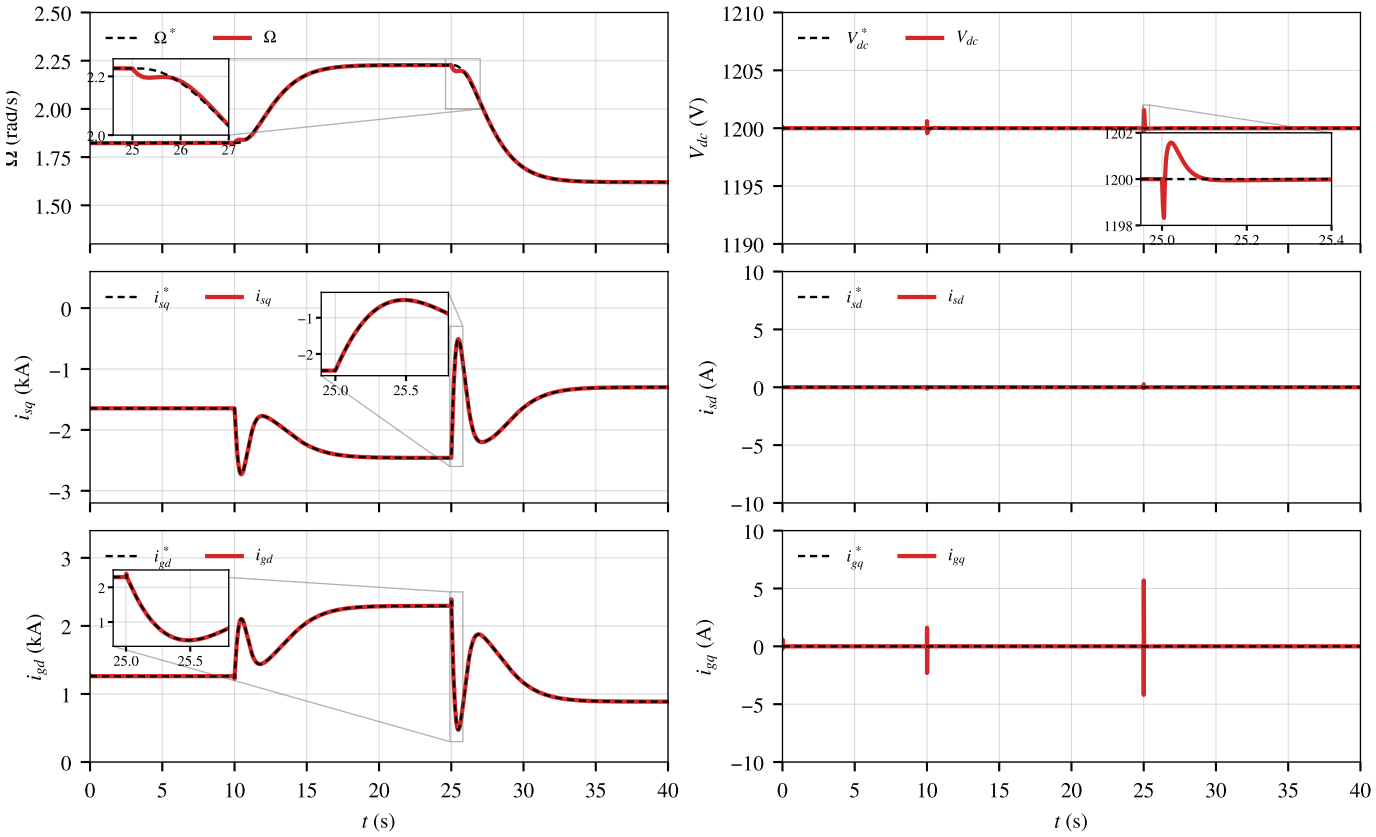


Fig. 4. Step wind, control with the RBF network: references (dashed) and measured values (solid) of the rotor speed, DC-link voltage and stator and grid currents, with zooms on the step at 25 s.

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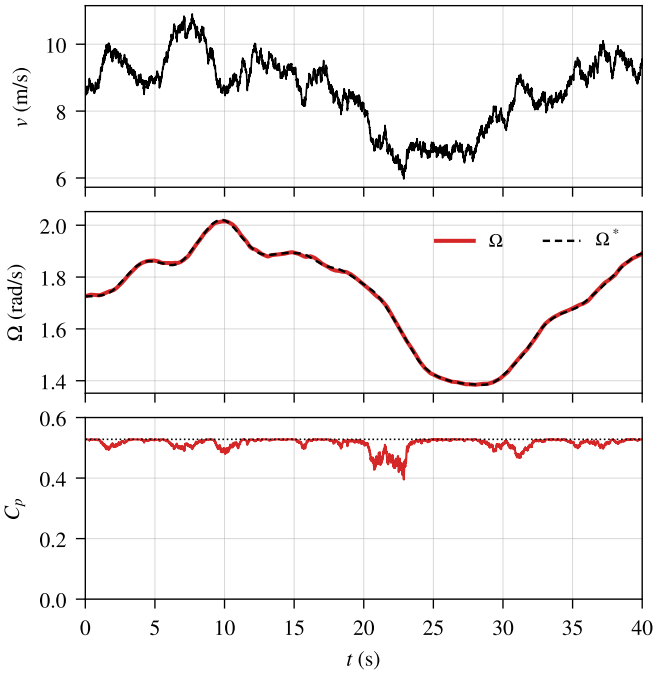


Fig. 5. Robustness case with the RBF network: wind speed, rotor speed and reference, and power coefficient (dotted: $C_{p,\max}$ of the true turbine).

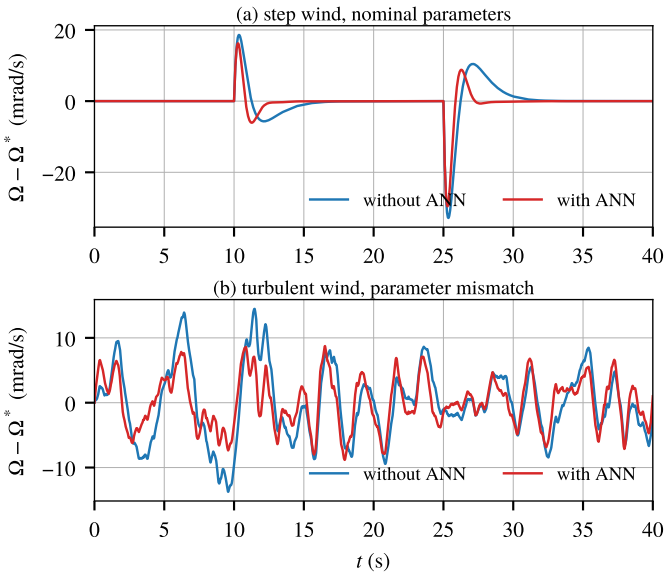


Fig. 6. Speed tracking error without and with the RBF network: (a) step wind; (b) robustness case.

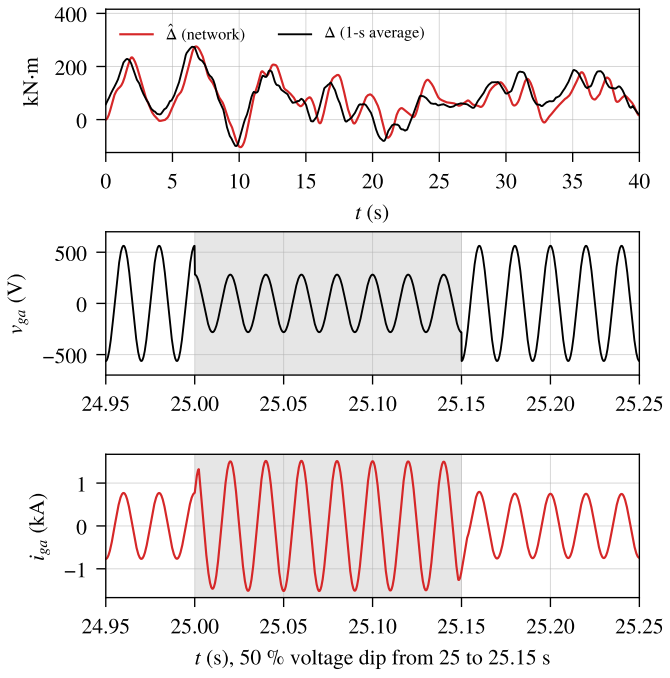


Fig. 7. Robustness case: network output $\hat{\Delta}$ and actual uncertainty Δ (1-s average), and grid voltage and current of phase a during the 50 % voltage dip.